

ANIMALCARE HUB

Animal Rescue, Veterinary Care, Vaccination & Adoption Management System

1. STUDENT DETAILS

Student Name: Pranay Jagdish Munde

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Division: B

Project Title: AnimalCare Hub – Animal Rescue, Veterinary Care,
Vaccination & Adoption Management System

2. SELECTED PROBLEM STATEMENT

Animal rescue organizations and veterinary care centers need to maintain records of rescued animals, veterinarians, treatments, vaccinations, and adoption requests.

Managing these records manually can make searching, updating, and maintaining information difficult.

AnimalCare Hub is a database-driven web application developed to provide a centralized system for managing animal rescue, veterinary care, vaccination, and adoption records.

3. OBJECTIVES AND MAIN FEATURES

Objectives:

1. To manage animal rescue, veterinary, vaccination, and adoption records.
2. To provide CRUD, search, and filtering operations for quick record management.
3. To track animal health, treatments, and vaccination schedules.
4. To provide secure database operations using prepared statements.
5. To deploy the application as a PHP-MySQL web application on InfinityFree.

Main Features:

- Animal Management – Add, view, update, search, and delete animals.
- Veterinary Management – Manage veterinarian and clinic details.
- Treatment Management – Record animal diagnosis and treatment.
- Vaccination Management – Record vaccines and next due dates.
- Adoption Management – Submit and update adoption requests.
- Search & Filtering – Search animal records easily.
- User Authentication – Registration, login, logout, and password hashing.
- Live Dashboard – Displays current database statistics.

4. TECHNOLOGY STACK

Frontend:

- HTML5
- CSS3
- JavaScript

Backend:

- PHP

Database:

- MySQL

Hosting:

- InfinityFree

5. DATABASE TABLE DESIGN

The application uses six main tables.

A. ANIMALS

Column Name	Type	Key / Constraints
animal_id	INT	Primary Key, Auto Increment
animal_name	VARCHAR(100)	NOT NULL
animal_type	VARCHAR(50)	NOT NULL
breed	VARCHAR(100)	-
age	INT	NOT NULL
gender	VARCHAR(20)	NOT NULL
category	VARCHAR(30)	NOT NULL
rescue_date	DATE	-
health_status	VARCHAR(50)	NOT NULL
adoption_status	VARCHAR(30)	NOT NULL
description	TEXT	-

B. VETERINARIANS

Column Name	Type	Key / Constraints
vet_id	INT	Primary Key, Auto Increment
vet_name	VARCHAR(100)	NOT NULL
specialization	VARCHAR(100)	NOT NULL
phone	VARCHAR(20)	-
email	VARCHAR(150)	UNIQUE
clinic_name	VARCHAR(150)	-

C. TREATMENTS

Column Name	Type	Key / Constraints
treatment_id	INT	Primary Key, Auto Increment
animal_id	INT	Foreign Key
vet_id	INT	Foreign Key
treatment_date	DATE	NOT NULL
diagnosis	VARCHAR(255)	NOT NULL
treatment_description	TEXT	NOT NULL
treatment_status	VARCHAR(30)	NOT NULL

D. VACCINATIONS

Column Name	Type	Key / Constraints
vaccination_id	INT	Primary Key, Auto Increment
animal_id	INT	Foreign Key
vaccine_name	VARCHAR(100)	NOT NULL
vaccination_date	DATE	NOT NULL
next_due_date	DATE	NOT NULL
vaccination_status	VARCHAR(30)	NOT NULL

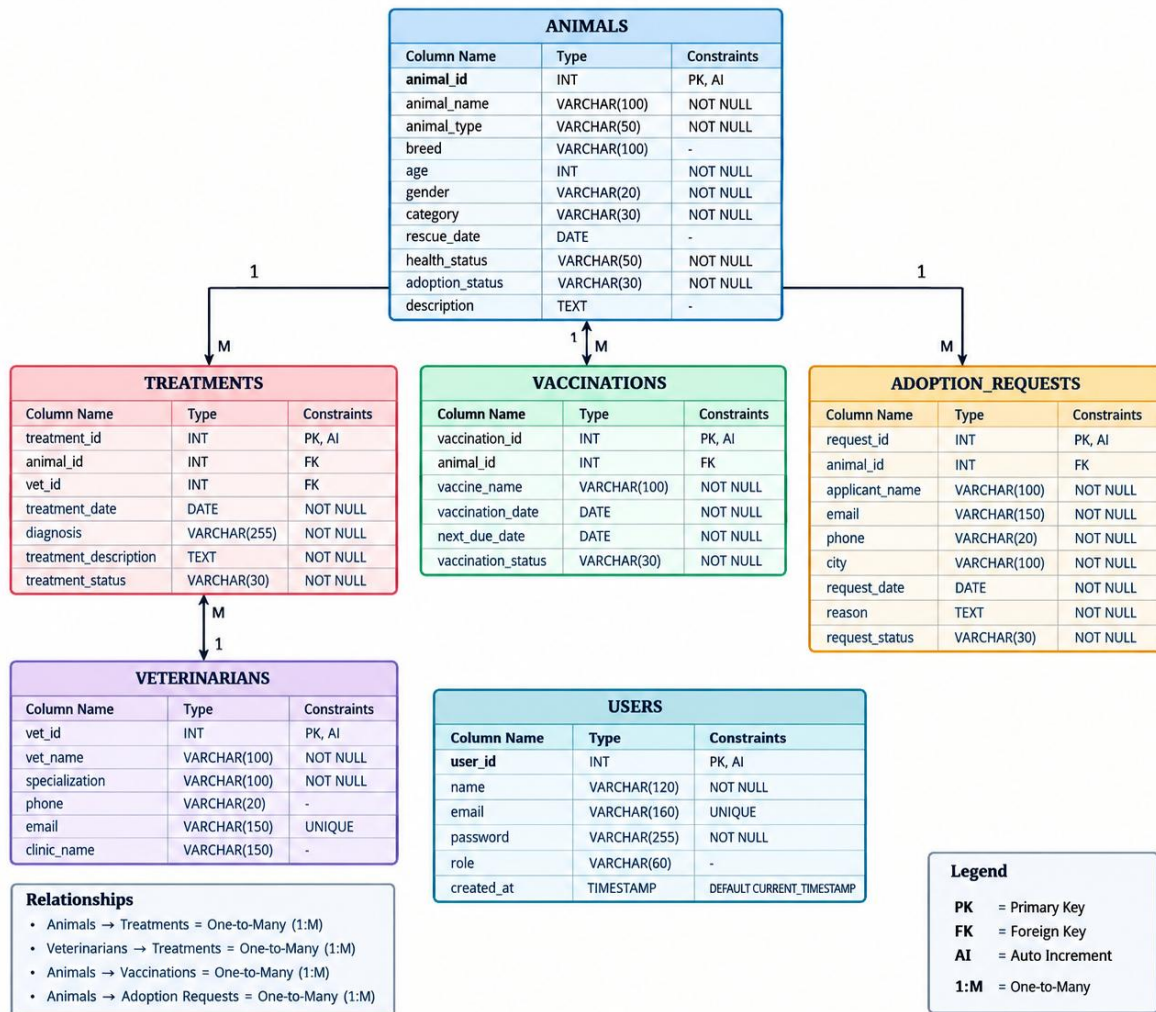
E. ADOPTION_REQUESTS

Column Name	Type	Key / Constraints
request_id	INT	Primary Key, Auto Increment
animal_id	INT	Foreign Key
applicant_name	VARCHAR(100)	NOT NULL
email	VARCHAR(150)	NOT NULL
phone	VARCHAR(20)	NOT NULL
city	VARCHAR(100)	NOT NULL
request_date	DATE	NOT NULL
reason	TEXT	NOT NULL
request_status	VARCHAR(30)	NOT NULL

F. USERS

Column Name	Type	Key / Constraints
user_id	INT	Primary Key, Auto Increment
name	VARCHAR(120)	NOT NULL
email	VARCHAR(160)	UNIQUE
password	VARCHAR(255)	NOT NULL
role	VARCHAR(60)	-
created_at	TIMESTAMP	Default Current Timestamp

6. ER DIAGRAM / TABLE SCHEMA



Relationships:

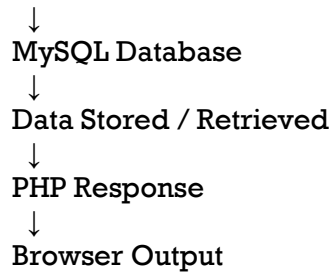
- Animals → Treatments = One-to-Many
- Veterinarians → Treatments = One-to-Many
- Animals → Vaccinations = One-to-Many
- Animals → Adoption Requests = One-to-Many

7. APPLICATION FLOW

The application follows this flow:

```

User
↓
HTML Form
↓
Client/Server Validation
↓
PHP Backend
↓
Prepared SQL Statement
    
```



Process:

1. The user enters data through an HTML form.
2. Form data is validated.
3. PHP receives the submitted data.
4. PHP establishes a connection with MySQL.
5. A prepared SQL statement is created.
6. User input is passed using parameter binding.
7. MySQL performs the required operation.
8. PHP retrieves the result.
9. The result is displayed in the browser.

8. SQL OPERATIONS USED

The project demonstrates the following SQL operations:

INSERT:

Used to add new animal, veterinarian, treatment, vaccination, and adoption records.

SELECT:

Used to retrieve and display stored records.

SEARCH:

Used to search and filter animal records based on user input.

UPDATE:

Used to modify existing records.

DELETE:

Used to remove records safely.

JOIN:

INNER JOIN is used to retrieve related information from multiple tables.

Other DBMS concepts used:

- CREATE TABLE
- PRIMARY KEY
- FOREIGN KEY
- NOT NULL
- UNIQUE
- AUTO_INCREMENT

- Transactions

9. PREPARED STATEMENTS

Prepared statements are used to protect the application from SQL injection attacks.

Instead of directly joining user input with an SQL query, the application uses a `?` placeholder and binds the user input as a parameter.

Example:

```
$stmt = $conn->prepare(  
    "SELECT * FROM animals WHERE animal_name LIKE ?"  
);
```

```
$stmt->bind_param("s", $search);
```

This ensures that user input is treated as data rather than as part of the SQL command.

Prepared statements are used for user-input database operations such as INSERT, SELECT, SEARCH, UPDATE, and DELETE.

10. HOSTED APPLICATION URL

Hosting Platform: InfinityFree

Hosted URL:

<https://animalcarehub.rf.gd/index.php>

11. TEST CASES AND OUTPUT SCREENSHOTS

The application was tested using the following test cases:

Test Case 1: Add Animal

Input: Valid animal details

Expected Result: Animal record should be inserted.

Result: PASS

Test Case 2: View Records

Input: Open animal records page

Expected Result: Saved records should be displayed.

Result: PASS

Test Case 3: Search Animal

Input: Search/filter value

Expected Result: Matching records should be displayed.

Result: PASS

Test Case 4: Update Animal

Input: Modified animal details

Expected Result: Existing record should be updated.

Result: PASS

Test Case 5: Delete Animal

Input: Delete request

Expected Result: Selected record should be removed safely.

Result: PASS

Test Case 6: Security Test

Input: O'Reilly

Expected Result: Value should be stored/displayed without breaking the SQL query.

Result: PASS

Test Case 7: Hosted Application

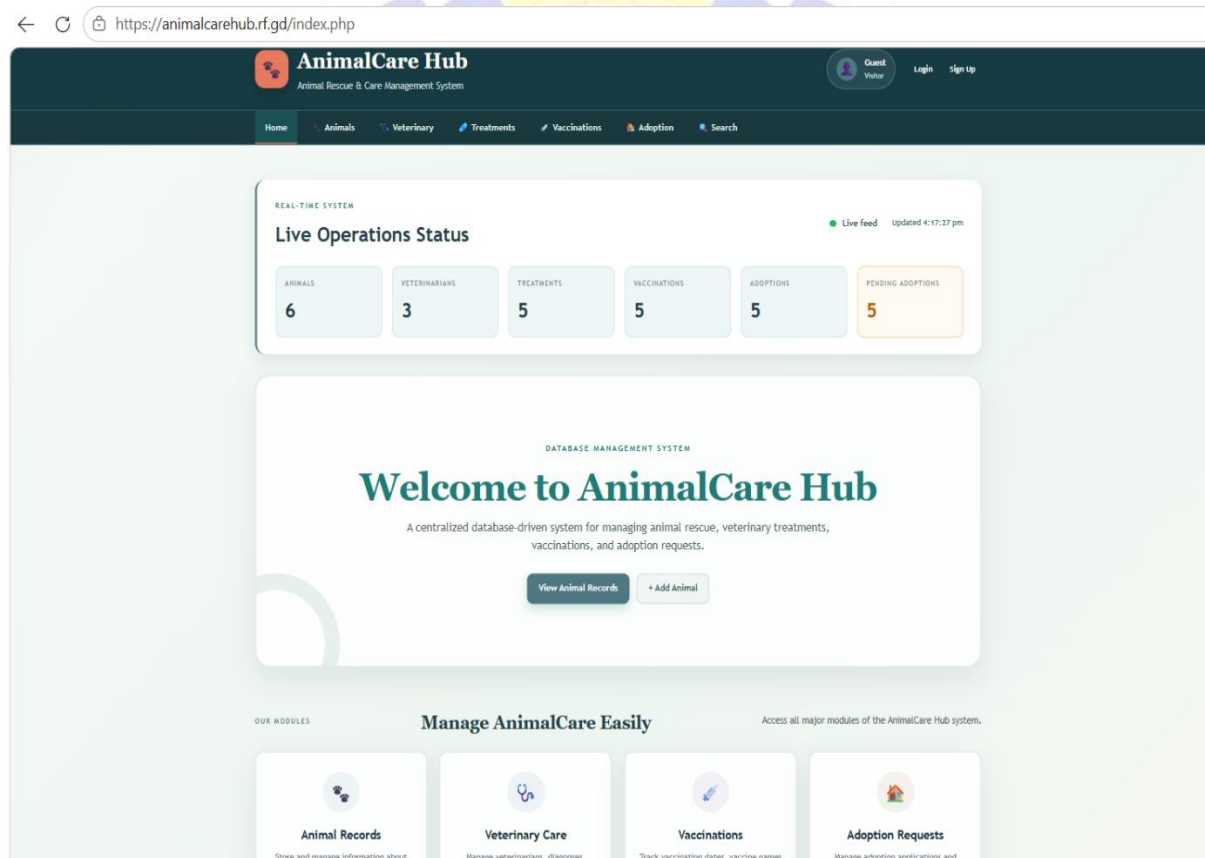
Input: Open InfinityFree URL

Expected Result: Application should load successfully.

Result: PASS

OUTPUT SCREENSHOTS

Screenshot 1 – Home / Data Entry Form



Database Management System

Screenshot 2 – Completed Form Before Submission

ANIMAL DATABASE
Add New Animal
Enter the animal information and save it to MySQL.

Basic Information

Animal Name *
Milky

Animal Type *
Dog

Breed
Labrador

Age *
2

Gender *
Male

Category *
Pet

Health & Adoption

Rescue Date
10-09-2026

Health Status *
Healthy

Adoption Status *
Available

Description
White coloured cute dog.

Cancel Reset Save Animal

Screenshot 3 – Successful Insert Confirmation

✓ Animal record added successfully.

ANIMAL DATABASE
Animal Records
Add, edit, search and safely manage all registered animal records.

RECORDS
All Animals

Search Add Animal

ID	NAME	TYPE	BREED	AGE	GENDER	CATEGORY	HEALTH	ADOPTION	ACTIONS
15	Jul	Dog	German Shepherd	1	Female	Pet	Healthy	Available	Edit Delete
14	Milky	Dog	Labrador	2	Male	Pet	Healthy	Available	Edit Delete
13	Tom	Cat	Persian	2	Male	Shelter	Recovered	Pending	Edit Delete
11	Reva	Dog	German Shepherd	4	Male	Rescue	Healthy	Available	Edit Delete
9	Lucy	Dog	Labrador	5	Female	Pet	Recovered	Adopted	Edit Delete
5	Rocky	Dog	German Shepherd	5	Male	Rescue	Recovering	Available	Edit Delete
4	Bella	Rabbit	Angora	1	Female	Rescue	Healthy	Adopted	Edit Delete
3	Coco	Dog	Indie	4	Female	Pet	Healthy	Available	Edit Delete
2	Milo	Cat	Persian	2	Male	Rescue	Healthy	Available	Edit Delete
1	Bruno	Dog	Labrador	3	Male	Rescue	Healthy	Available	Edit Delete

Database Management System

Screenshot 4 – Record List Showing Saved Data

The screenshot displays the 'AnimalCare Hub' interface. At the top, there's a navigation bar with 'Home', 'Animals', 'Veterinary', 'Treatments', 'Vaccinations', 'Adoption', and 'Search'. A 'Guest Visitor' button is also present. Below the navigation bar, a green notification box states '✓ Animal record added successfully.' The main section is titled 'ANIMAL DATABASE' and 'Animal Records', with a sub-header 'Add, edit, search and safely manage all registered animal records.' A 'RECORDS' section shows a table of 'All Animals' with columns: ID, NAME, TYPE, BREED, AGE, GENDER, CATEGORY, HEALTH, ADOPTION, and ACTIONS. The table lists 15 animals, including Jui, Milky, Tom, Reva, Lucy, Rocky, Bella, Coco, Milo, and Bruno. Each row has 'Edit' and 'Delete' buttons. A search bar and '+ Add Animal' button are at the top right of the table.

ID	NAME	TYPE	BREED	AGE	GENDER	CATEGORY	HEALTH	ADOPTION	ACTIONS
15	Jui	Dog	German Sheopherd	1	Female	Pet	Healthy	Available	Edit Delete
14	Milky	Dog	Labrador	2	Male	Pet	Healthy	Available	Edit Delete
13	Tom	Cat	Persian	2	Male	Shelter	Recovered	Pending	Edit Delete
11	Reva	Dog	German Sheopherd	4	Male	Rescue	Healthy	Available	Edit Delete
9	Lucy	Dog	Labrador	5	Female	Pet	Recovered	Adopted	Edit Delete
5	Rocky	Dog	German Shepherd	5	Male	Rescue	Recovering	Available	Edit Delete
4	Bella	Rabbit	Angora	1	Female	Rescue	Healthy	Adopted	Edit Delete
3	Coco	Dog	Indie	4	Female	Pet	Healthy	Available	Edit Delete
2	Milo	Cat	Persian	2	Male	Rescue	Healthy	Available	Edit Delete
1	Bruno	Dog	Labrador	3	Male	Rescue	Healthy	Available	Edit Delete

Screenshot 5 – Search / Filter Working

The screenshot displays the 'AnimalCare Hub' interface with the 'Search' tab selected. The main section is titled 'ANIMAL DATABASE' and 'Search & Filter Animals', with a sub-header 'Search by animal name or breed and narrow results using the filters below.' Below this, there are four filters: 'Name / Breed' (text input with 'Bruno'), 'Animal Type' (dropdown with 'All Types'), 'Health Status' (dropdown with 'All Health Status'), and 'Adoption Status' (dropdown with 'All Adoption Status'). A 'Search' button and a 'Clear' button are at the bottom right of the filter section. The 'RESULTS' section shows '1 matching record' and a table with columns: ID, NAME, TYPE, BREED, AGE, GENDER, CATEGORY, HEALTH, ADOPTION, and ACTIONS. The table lists one animal, Bruno, with 'Edit' and 'Delete' buttons. A '+ Add Animal' button is at the top right of the results section.

ID	NAME	TYPE	BREED	AGE	GENDER	CATEGORY	HEALTH	ADOPTION	ACTIONS
1	Bruno	Dog	Labrador	3	Male	Rescue	Healthy	Available	Edit Delete

Database Management System

Screenshot 6 – Security Test using O'Reilly

The screenshot shows the 'Add New Animal' form in the AnimalCare Hub system. The form is titled 'ANIMAL DATABASE' and 'Add New Animal'. Below the title, it says 'Enter the animal information and save it to MySQL'. The form is divided into two main sections: 'Basic Information' and 'Health & Adoption'. In the 'Basic Information' section, the 'Animal Name' is 'O'Reilly', 'Animal Type' is 'Bird', 'Breed' is 'African', 'Age' is '2', 'Gender' is 'Female', and 'Category' is 'Rescue'. In the 'Health & Adoption' section, the 'Rescue Date' is '04-09-2026', 'Health Status' is 'Healthy', and 'Adoption Status' is 'Available'. There is a 'Description' field with the text 'Testing'. At the bottom of the form, there are three buttons: 'Cancel', 'Reset', and 'Save Animal'.

Basic Information	
Animal Name *	O'Reilly
Animal Type *	Bird
Breed	African
Age *	2
Gender *	Female
Category *	Rescue

Health & Adoption	
Rescue Date	04-09-2026
Health Status *	Healthy
Adoption Status *	Available
Description	Testing

Screenshot 7 – InfinityFree Hosted Application

The screenshot shows the homepage of the AnimalCare Hub system. The header includes the 'AnimalCare Hub' logo and navigation links for 'Home', 'Animals', 'Veterinary', 'Treatments', 'Vaccinations', 'Adoption', and 'Search'. The main content area features a 'REAL-TIME SYSTEM' section with 'Live Operations Status' and a 'Live feed' indicator. Below this, there are six cards showing counts for 'ANIMALS' (6), 'VETERINARIANS' (3), 'TREATMENTS' (5), 'VACCINATIONS' (5), 'ADOPTIONS' (5), and 'PENDING ADOPTIONS' (5). A 'Welcome to AnimalCare Hub' section follows, describing the system as a centralized database-driven system for managing animal rescue, veterinary treatments, vaccinations, and adoption requests. It includes buttons for 'View Animal Records' and '+ Add Animal'. At the bottom, there is a 'Manage AnimalCare Easily' section with four modules: 'Animal Records', 'Veterinary Care', 'Vaccinations', and 'Adoption Requests'.

REAL-TIME SYSTEM

Live Operations Status Live feed updated 4:17:37 pm

ANIMALS	VETERINARIANS	TREATMENTS	VACCINATIONS	ADOPTIONS	PENDING ADOPTIONS
6	3	5	5	5	5

WELCOME TO ANIMALCARE HUB

A centralized database-driven system for managing animal rescue, veterinary treatments, vaccinations, and adoption requests.

[View Animal Records](#) [+ Add Animal](#)

OUR MODULES

Manage AnimalCare Easily

Access all major modules of the AnimalCare Hub system.

Animal Records	Veterinary Care	Vaccinations	Adoption Requests
Store and manage information about	Manage veterinarians, diagnoses,	Track vaccination dates, vaccine names	Manage adoption applications and

12. DECLARATION OF ORIGINAL WORK

I hereby declare that the project titled

"AnimalCare Hub – Animal Rescue, Veterinary Care, Vaccination & Adoption Management System"

has been developed by me for academic purposes.

The project demonstrates my understanding of HTML, CSS, JavaScript, PHP, MySQL, database management concepts, SQL operations, database connectivity, prepared statements, and web application development.

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Signature: Pranay J. Munde

Date: 14th September 2026

